

1)The term handover in cellular networks refers to:

- a. The movement of a mobile terminal from a cell to another without the interruption of an active communication (e.g., a call)
- b. The movement of a mobile terminal from a cell to another, even if in that moment there are no active communications
- c. The turn-on of the mobile terminal after a long period of inactivity
- d. The use of a mobile terminal in the network of a different operator with respect to the one the SIM card installed in the terminal belongs to

2)With respect to previous solutions, The Long Term Evolution (LTE) technology is characterized, among the other things, by:

- a. The use of switching technique based on virtual circuits
- b. The separation between physical and logical channels
- c. The use of a circuit switching technique
- d. The use of IP tunnels connecting terminals with elements of the EPC

3)An optical switch is a device capable of switching:

- a. Different optical channels on a given wavelength on different input fibers to a given wavelength on a given output fiber.
- b. Packets arriving on a given optical channel from a given input fiber to different optical channels to different output fibers based on information in the header of each packet.
- c. An optical channel on a given wavelength from a given input fiber to different wavelengths on different output fibers.
- d. An optical channel on a given wavelength from a given input fiber to a given output fiber.

4)An IPSec-based VPN (Virtual Private Network)

- a. The encryption is mandatory
- b. Is based on a specific cryptographic algorithm defined by the IPsec standard itself.
- c. Deploys tunneling to allow encryption and/or authentication of IP packets exchanged.
- d. Requires that all communicating hosts support the IPsec protocol.

5)The shortcut IPv6 address 1070:5::1:2A4:E8:7782:BB59 corresponds to which of the following full-length addresses?

- a. 1070:5001:0000:0000:02A4:E008:7782:BB59
- b. 1070:0005:0000:0001:2A40:E008:7782:BB59

- c. 1070:5000:0000:1000:2A40:E800:7782:BB59
- d. 1070:0005:0000:0001:02A4:00E8:7782:BB59

6) A packet with destination IP address 180.3.13.223 will have a longest match in which of the following lines belonging to a routing table?

- a. 180.3.13.224/28
- b. None of the proposed solutions
- c. 180.3.13.208/29
- d. 180.3.13.192/28

7) The Solicited Node Multicast Address:

- a. Is a special multicast group that can contain one or more nodes. These nodes receive and process the ICMP Neighbor Solicitation, but only one will reply with an ICMP Neighbor Advertisement.
- b. Is a special multicast group that contains all nodes of a given Link, as it replaces the broadcast address in the IPv6 context. All nodes receive and process the ICMP Neighbor Solicitation, but only one will reply with an ICMP Neighbor Advertisement.
- c. Is an obsolete Neighbor Discovery technique, now replaced by a more efficient solution based on ARP.
- d. Is a special multicast group that contains only one node. This node receives and processes the ICMP Neighbor Solicitation, and then it will reply with an ICMP Neighbor Advertisement.

8) OSPF-TE and ISIS-TE include enhancements to traditional routing protocols OSPF and ISIS that have been standardized in the context of MPLS in order to:

- a. Allow the independence of control plane and data plane in MPLS routers, which ultimately enables traffic engineering.
- b. Ensure faster convergence and enhanced stability.
- c. Enable the distribution of additional information (named constrained data) in support of constraint based routing.
- d. Include adequate fields in routing messages to support explicit routing.

9) The IPv6 Aggregatable Global Unicast addresses are:

- a. Globally unique, substantially equivalent to the IPv4 public addresses
- b. Aggregatable only in very small address ranges, in order to favor the precision of the evaluation of routing paths
- c. Usable in a global IPv6 network only with proper Network Address Translation (NAT) techniques

d. Usable only on devices belonging to the same local area network

10)The smallest aggregation that can contain the IP networks 130.192.0.0/25 and 130.192.1.0/25 is

- a. 130.192.1.0/23
- b. 130.192.0.0/24
- c. 130.192.0.0/23
- d. 130.192.1.0/24

11)The IPv6 address 2001:4600::0201:06FF:FEA5:3A4C is:

- a. An address that can be used on a server to offer a service over the public IPv6 Internet
- b. An address currently not valid in IPv6
- c. An address that can be used by a host only to perform communications with another host on the same link
- d. A private address

12)Which of the following choices is not an Adaptive Routing algorithm?

- a. Isolated Routing
- b. Link State
- c. Centralized Routing
- d. Selective Flooding

13)One of the reasons that are favoring the spread of the IPv6 protocol is:

- a. The possible inefficiency of the private IPv4 addressing
- b. The low inclination of network operators to modify the configuration of their own networks
- c. The lack of MAC addresses
- d. The more and more widespread need to use multicast applications

14)In order to setup a label switched path (LSP):

- a. MPLS routers on the path must identify a label that is unused on all links of the path.
- b. MPLS routers on the path must perform a mapping operation
- c. Final destinations of IP packets traveling on the LSP must support MPLS
- d. The same layer-two protocol must be deployed on all links on the path

15)The adoption of the MAC address as Interface ID of an IPv6 host:

- a. allows the host to perform a direct communication without the need to insert a Layer 2 header
- b. is not possible due to privacy issues
- c. is part of one of the possible autoconfiguration solutions available in IPv6
- d. is the only possible solution available in IPv6 for the Interface ID configuration

16)One of the functions of the IPv4 protocol suite that was replaced by ICMPv6 is:

- a. IPsec
- b. PPP
- c. OSPF
- d. IGMP

17)The static routing:

- a. Consists in the automatic learning of routes without exchanging routing information
- b. Consists in one network node computing routes for other network nodes and providing the computed routes to them
- c. Consists in the network administrator manually configuring routing information
- d. It is an obsolete technology no longer deployed since dynamic routing is preferred

18)The configuration of a switch port in access mode is used for:

- a. Assigning a packet sent through that port to a specific VLAN
- b. Assigning a packet received on that port to a specific VLAN
- c. Enabling the access to the network
- d. Allowing the access to the switch only to specific flows that are not considered

19)The IPv6 Extension Headers are:

- a. Header chains that can be added to the main IPv6 header in order to move to the network some features that are typically of the transport layer (e.g., the transmission of acknowledgments)
- b. Padding techniques adopted to make the IPv6 packet of fixed size equal to 40 bytes
- c. Header chains that can be added to the main IPv6 header in order to offer additional
- d. Padding techniques adopted to fix the size of the layer 2 frame containing the IPv6

20)The GRE (Generic Routing Encapsulation) protocol is utilized for:

- a. Encapsulation of any protocol into an IP packet
- b. Exchanging routing information in various protocol architectures
- c. Multiple transmission operations
- d. Implementing encrypted tunneling in secure VPNs (Virtual Private Networks)

21)A Layer2 Ethernet switch is used for interconnecting 47 hosts belonging to three different VLANs: 20 hosts belong to VLAN 2, 20 hosts belong to VLAN 3, 17 hosts belong to VLAN 4. All the 47 ports of the switch must be in Access mode.

- a. It is not possible to enable the communication among all the hosts because at least two available ports would be needed
- b. It is possible to enable the communication among all the hosts using the single port that is still available in addition to proper network devices
- c. It is not possible to enable the communication among all the hosts because, being them connected to three VLANs at least three available ports would be needed
- d. The 47 hosts can communicate without any further change to the network configuration

22)In LTE networks the use of tunnels facilitates:

- a. the encryption of traffic between operators
- b. the connectivity between operators
- c. the compression of control packet headers
- d. the mobility of users between cells

23)Why is label-based forwarding less complex (and hence can be implemented using simpler hardware with lower power consumption) than IP address-based forwarding?

- a. Because label's validity is local to a link, while IP address's validity is global across the network
- b. Because label-based forwarding can be implemented with an exact match in a table, while address-based forwarding requires a longest prefix match
- c. Because labels can be chosen by routers, while IP addresses must be chosen by network administrators
- d. Because labels don't need to be unique, while IP addresses must be unique across the network (unless NAT is deployed)

24)What is a typical operation of an all-optical switch?

- a. Process and switch individual bits transmitted on wavelengths belonging to the optical portion of the spectrum
- b. Route an optical channel (at a specific frequency) from an input port to an output port
- c. Switch packets received from an optical channel to different optical channels
- d. Multiplex over a single outgoing optical channel data received over different incoming channels

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26)When an MPLS frame has multiple labels, routers:

- a. Only process the most external label to decide how to forward the frame
- b. Only can swap the most external and the next label with each other
- c. Only process the most internal label to decide how to forward the frame
- d. Can swap any of the labels

27)In IEEE 802.11 networks, the "hidden terminal problem" refers to:

- a. a security threat caused by an intruder in the BSS
- b. the use of the same MAC address by more than one node in the same BSS
- c. the possibility that two or more wireless nodes can interfere with each other without
- d. a failure in the association to an Access Point during the passive scanning procedure

28)Concerning the DiffServ (Differentiated Services) standard and practice of traffic engineering:

- a. they both rely on information included in the IP header
- b. they are completely independent and unrelated solutions that are never used in conjunction
- c. they are two different tools to do the same

d. they can be used in conjunction to perform per-class traffic engineering, which enables efficient utilization of network capacity

29) Multicast communications in an IPv4 network:

- a. Are not possible without the deployment of additional protocols in the network
- b. Are possible at global scale, it is only required to enable the IGMP protocol in the network
- c. Are always possible, IGMP only makes them more efficient
- d. Are possible only within a single LAN, even if additional protocols are used

30) Which of the following methods is used by Content Delivery Networks (CDNs) to handle content updates efficiently while minimizing disruptions to users?

- a. Multi-path TCP (MPTCP): Utilizing multiple TCP connections simultaneously to enhance performance and reliability between clients and origin servers
- b. Compression Algorithms: Applying real-time compression to reduce the size of content from the origin server
- c. Edge Caching: Storing frequently accessed content on servers located closer to end users with cache invalidation policies to ensure content freshness
- d. Dynamic Content Acceleration: Optimizing the delivery of dynamically generated content by bypassing the CDN cache and fetching content directly from the origin server

31) When MPLS was standardized, what was the main motivation for having labels pushed into packets by network nodes (specifically by Label Edge Routers) rather than end hosts?

- a. End hosts are less trusted than network nodes because they can be more easily hacked
- b. End hosts don't need to be aware of MPLS, which made the introduction of MPLS less problematic
- c. Pushing the label requires modifying the layer two header that is not present when a packet is transmitted and received by an end host
- d. Network nodes have higher performance than end hosts

32) A network interface with both an IPv4 and an IPv6 address can receive:

- a. Only IPv6 packets encapsulated in IPv4, according to the approach called "tunneling"
- b. Only IPv4 packets encapsulated in IPv6, according to the approach called "tunneling"
- c. Both IPv4 and IPv6 packets
- d. Only IPv6 packets, since in such kind of configuration the operating system considers IPv4 as obsolete

33) Given a network based on several physical networks interconnected by routers and a range of IP addresses to use in that network, it is possible to define an addressing plan that optimizes routing on a given router of the network by:

- a. Assigning to the various physical networks distinct network IDs randomly selected within the address range given for the entire network
- b. Assigning to the various physical networks distinct network IDs selected within the address range given for the entire network. In particular, this assignment must proceed from in a decreasing order of network size
- c. Splitting the network in areas and defining, within the given address range, smaller distinct address ranges to use in each area